Groq vs RunPod — LLM Strategy Decision

Project: ShiftED AI / Empathy Coach | **Date:** June 2026 | **Version:** 1.0

2-minute executive summary

	Groq (hosted API)	RunPod (fine-tuned)
Ease	Much easier — API key only	Harder — GPU, volume, cold start
Idle cost	**$0**	$0 at scale-to-zero; ~$7–11/day if warm 10h
Demo month	**~$1–7** (70B, moderate use)	~$15–40 (zero scale) or ~$150–240 (warm)
Custom model	No	**Yes — empathy-coach-qwen**
Best for	Demos, reliability, low ops	Long-term moat, Simon-trained tone

Recommendation: Use **Groq for near-term demos**; keep **RunPod fine-tune** as the strategic path. Hybrid switch via Netlify env vars only.

1. What we have today

RunPod (current target)

- Base: **Qwen2.5-7B-Instruct**, LoRA-trained on Simon feedback (~221 turns)
- Model: `empathy-coach-qwen` on volume `empathy-coach-qwen-merged-v1`
- RunPod Serverless vLLM, scale to zero, async warm-up UI in app

Groq (alternative)

- Hosted models only (e.g. `llama-3.3-70b-versatile`) — cannot deploy our fine-tune
- Same super prompt stack (phases, journey, trainer rules, memory)

2. Pros and cons

Groq — Pros

- Instant inference, no cold start
- Pay per token — $0 when idle
- Very low cost at demo scale (~$0.01–$2/day typical)
- Simple ops: API key + env vars
- Free tier for prototyping

Groq — Cons

- Cannot run empathy-coach-qwen fine-tune
- Quality depends on base model + prompt only
- Rate limits on free tier; card required for production
- User coaching data sent to Groq infrastructure

RunPod — Pros

- Our fine-tuned model trained on Simon feedback
- Re-trainable on new Supabase feedback
- Scale to zero — no cost when idle
- EU volume, owned weights

RunPod — Cons

- Operational complexity (endpoint, GPUs, logs)
- Cold start 1–3 min after idle
- GPU availability issues (e.g. 80 GB unavailable)
- Warm-up for instant demos costs ~$7–11/day

3. Cost comparison

Groq bills per token. RunPod bills per GPU-second while a worker runs. "10 hours/day" applies to RunPod warm-up, not Groq.

Groq pricing (June 2026)

Model	Input / 1M	Output / 1M
Llama 3.1 8B Instant	$0.05	$0.08
Llama 3.3 70B Versatile	$0.59	$0.79
Qwen3 32B	$0.29	$0.59

Groq — estimated monthly (weekdays)

Scenario	Chats/day	8B	70B
Light demo	~20	~$0.20/mo	~$1.30/mo
Team testing	~100	~$0.70/mo	~$7/mo
Active pilots	~500	~$3/mo	~$35/mo

RunPod — estimated monthly

Mode	RTX 4090 (~$1.10/hr)	L4 (~$0.68/hr)
Scale-to-zero, light use	~$10–40/mo	~$5–20/mo
Warm 10h/day (cron)	~$240/mo	~$150/mo
Min workers = 1 (24/7)	~$790/mo	~$490/mo

4. Decision matrix

Priority	Choose
Reliability for demos this week	**Groq**
Lowest operational burden	**Groq**
Simon-aligned coaching from weights	**RunPod fine-tune**
Own model IP / re-train on feedback	**RunPod fine-tune**

5. Switch checklists

Groq (Netlify)

```
VLLM_API_URL=https://api.groq.com/openai/v1/chat/completions
LLM_API_KEY=<Groq API key>
VLLM_MODEL=llama-3.3-70b-versatile
RUNPOD_ASYNC=false
```

RunPod (fine-tuned)

```
VLLM_API_URL=https://api.runpod.ai/v2/<ENDPOINT_ID>/openai/v1/chat/completions
LLM_API_KEY=<RunPod API key>
VLLM_MODEL=empathy-coach-qwen
RUNPOD_ASYNC=true
VLLM_MAX_CONTEXT_TOKENS=6144
```

6. Suggested hybrid roadmap

1. **Now:** Groq 70B for demos and stakeholder testing
2. **Parallel:** Stabilise RunPod (4090/L4, volume, logs)
3. **Next:** Simon A/B — fine-tuned RunPod vs Groq 70B + super prompt
4. **Decide:** Production path based on coaching quality review

7. References

- Groq: console.groq.com
- RunPod pricing: docs.runpod.io/serverless/pricing
- In-repo: `docs/SUPER-PROMPT-RUNPOD-OWN-LLM.md`, `docs/TRAIN-ON-SIMON-FEEDBACK.md`

Re-verify provider pricing before budget sign-off. Full markdown: `docs/GROQ-VS-RUNPOD-DECISION.md`